

B. Division Algorithm. In elementary school you learned the method of long division, which can assist us for example in executing $432 \div 15$.

Then: $[\text{dividend}] = [\text{quotient}] \cdot [\text{divisor}] + [\text{remainder}]$

in our case: $432 =$

Division Algorithm.

Let a be an integer (the **dividend**) and b be a positive integer (the **divisor**).

Then there exist **unique** integers q (the **quotient**) and r (the **remainder**) so:

$a =$

Proof. Long-division provides gives us a quotient and remainder.

So, there do exist integers q and r satisfying:

$a =$

Now, we need to prove the quotient and remainder are **unique**.

To this end, let q' and r' be integers satisfying:

$a =$

To confirm uniqueness of the quotient and remainder, we must show:

We subtract our two expressions for a to obtain the following.

$0 =$

We move terms to each side and take absolute values.

Because r and r' are in $[0 \dots b - 1]$, it follows that:

The only multiple of b in $[0 \dots b - 1]$ is:

Since $|r - r'|$ has been expressed as a positive multiple of b , it follows:

Or in other words:

Uniqueness is established.

□

Technically we should provide an explanation that the method of long-division concludes in finite time, and that what it concludes with is a quotient and remainder, in the sense that $a = bq + r$ and $0 \leq r < b$.

Example 3. Let n be an integer.

Prove that n^4 is either of the form $5k$ or $5k + 1$, where k denotes an integer.

Solution. By the division algorithm for $n \div 5$, we know n has form either:

$$n =$$

$$n =$$

$$n =$$

$$n =$$

$$n =$$

By the binomial theorem, we observe:

$$(5k + r)^4 = \binom{4}{0}r^4 + \binom{4}{1}(5k)^1 \cdot r^3 + \binom{4}{2}(5k)^2 \cdot r^2 + \binom{4}{3}(5k)^3 \cdot r^1 + \binom{4}{4}(5k)^4$$

and then noting that 5 may be factored out of every term but:

$$(5k + r)^4 =$$

We apply this observation to the different forms of n .

$$n^4 = (5k + 0)^4 =$$

$$n^4 = (5k + 1)^4 =$$

$$n^4 = (5k + 2)^4 =$$

$$n^4 = (5k + 3)^4 =$$

$$n^4 = (5k + 4)^4 =$$

C. **Divisibility.** We say integer a is **divisible** by integer b if and only if:

• $a =$

and we denote this by:

Read this as “ b divides a ”.

or alternatively say that b is a:

or that a is a:

Useful properties of divisibility are assembled in your exam **reference sheet**.

We establish a few here.

Let a, b, c, d be integers such that $a \mid c$ and $b \mid d$. Then $ab \mid cd$.

Proof. Because $a \mid c$ and $b \mid d$ it follows that:

$c =$

$d =$

Then:

$cd =$

is a multiple of ab . Thus $ab \mid cd$.

□

Let a and b be integers.

An **integral combination** of a and b refers to any integer of the form:

Let d be a **common divisor** of a and b .

Then, d is a divisor of any integral combination of a and b .

Proof. Let $ax + by$ be an integral combination of a and b , with x and y integers.

Because d is a common divisor of a and b we may write:

$a =$

$b =$

Then:

$ax + by =$

is a multiple of d . Thus $d \mid (ax + by)$. □